

AGENTIC CINEMA

Project Functionality Overview

An agentic AI production-planning system for turning screenplay PDFs into practical filmmaking plans.

1. Original Functionality

The prototype established the core screenplay-to-production workflow:

- Upload a screenplay PDF and extract its text.
- Use a Script Analysis Agent to identify the title, characters, locations, scenes, and story summary.
- Use a Production Planning Agent to determine shooting complexity, required locations, permits, lighting, production notes, and estimated shooting days.
- Use a Scheduling Agent to create a practical, location-based shooting schedule.
- Use a deterministic location information tool for production facts such as indoor or outdoor type, lighting, complexity, and permit requirements.
- Use a ClickHouse tool to retrieve stored production records for locations and scenes.
- Provide FastAPI, Streamlit, and React interfaces for the workflow.
- Display script analysis, production planning, and shooting schedule results in a dashboard.

2. Agentic System

The system is composed of specialized agents rather than one general chatbot:

Input	Agent or Tool	Output
Screenplay PDF	PDF Reader	Extracted screenplay text
Screenplay text	Script Analysis Agent	Characters, locations, scenes, summary
Script analysis	Production Planning Agent + Location Tool	Production requirements and notes
Analysis + production plan	Scheduling Agent	Shooting days and location schedule
Filmmaker question	ClickHouse Production Assistant	Database-backed recommendation

3. Functionality Added

- Typed validation schemas for script analysis, production plans, schedules, readiness results, and assistant requests.
- Reliable JSON parsing for Gemini responses, including Markdown-wrapped JSON.
- Demo mode for running the complete workflow without Gemini or ClickHouse credentials.
- Production Readiness Score out of 100.
- Readiness status such as Ready to schedule, Review before lock, or Needs production review.
- Risk Register for permits, high-complexity locations, missing scene details, and schedule mismatches.
- Recommended next production actions generated from the plan.
- AI Agent Execution Trace showing which specialized steps completed.
- Production Assistant connected to the React dashboard through the /assistant endpoint.
- Health monitoring through the /health endpoint.
- Export the complete production plan as JSON and print a production report.
- Backward-compatible frontend fallback when an older backend response does not yet include readiness data.
- Automated tests, Python dependency management, ClickHouse schema and seed data, and a testing screenplay PDF.

4. Current Dashboard Capabilities

- Screenplay upload and analysis workflow.
- Script title, summary, characters, locations, and scenes.
- Production complexity and estimated shooting days.
- Location requirements, permits, lighting, and production notes.
- Risk alerts and next actions.
- Agent activity trace.
- Day-by-day shooting schedule.
- AI production assistant for location and permit questions.
- JSON export and print report actions.
- Light and dark theme support.

5. End-to-End Workflow

Screenplay PDF -> PDF Reader -> Script Analysis Agent -> Production Planning Agent -> Location Tool / ClickHouse -> Scheduling Agent -> Readiness Evaluator -> Production Dashboard -> Production Assistant

6. Demonstration Mode

For competition demonstrations, the system can run locally without external services by setting:

AGENTIC_CINEMA_DEMO=1

This produces a complete sample analysis, production plan, schedule, readiness score, risk register, and assistant response.

Prepared for Agentic Cinema project presentation